Comparing Temporal Logic Reward Shaping Approaches in Reinforcement Learning

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1. Abstract

Temporal logic specifications are increasingly used to provide safety and task guarantees for reinforcement learning agents, yet most existing approaches assume infinite-horizon objectives. We empirically compare infinite-horizon automaton-based methods (**LTL**) with finite-horizon robustness-based formulations (**TLTL**) across three MiniGrid benchmark environments of increasing complexity, evaluating specification satisfaction probability, mean time to satisfaction, and sample efficiency across 30 independent runs. On simple tasks, all three methods performed comparably. On the sequential **DoorKey-8x8** task, **TLTL achieved 94.8% satisfaction vs 7.2% (LTL) and 0.6% (Baseline) ($p<0.001$)**, demonstrating that finite-horizon robustness rewards provide significantly more effective learning signals for complex ordered subtask problems, while infinite-horizon methods may offer stronger theoretical guarantees.

2. Background

- Reinforcement learning agents learn from scalar reward signals, which are difficult to design for complex, multi-step tasks [6].
- **Temporal logic** provides a formal language for expressing behavioral specifications like safety constraints, reachability goals, and ordered subtask completion [1].
- **Infinite-horizon LTL** encodes specifications as finite automata, providing *sparse rewards* upon reaching accepting states, which is effective but can suffer from slow convergence [4, 5].
- **Finite-horizon TLTL** computes a continuous *robustness value* measuring how well a trajectory satisfies the specification, providing *dense reward feedback* at every timestep [2, 3].
- Prior work has explored each approach independently, but direct empirical comparisons across task complexities remain limited.

3. Methods

Three reward shaping conditions (see Figure 1) were evaluated using **Proximal Policy Optimization (PPO)** [6] with an MLP policy across three **MiniGrid** environments of increasing complexity (see Figure 2).

- **Baseline:** Vanilla PPO with no temporal logic shaping. The agent receives only the environment's default sparse reward signal.
- **Infinite-Horizon LTL:** Specifications hand-coded as finite automaton wrappers tracking states (*Q_running, Q_accept, Q_dead*), providing *+1 reward on acceptance* and *-1 on violation* [2, 4].
- **Finite-Horizon TLTL:** Specifications hand-coded as robustness functions computing a *continuous satisfaction degree* at each timestep. The shaped reward is the change in robustness, providing *dense feedback* throughout the episode [2, 3].

All conditions used identical hyperparameters (100,000 timesteps, learning rate 3×10^{-4}, MlpPolicy) to ensure fair comparison, trained across 30 independent runs and evaluated over 100 episodes per run.

Figure 1

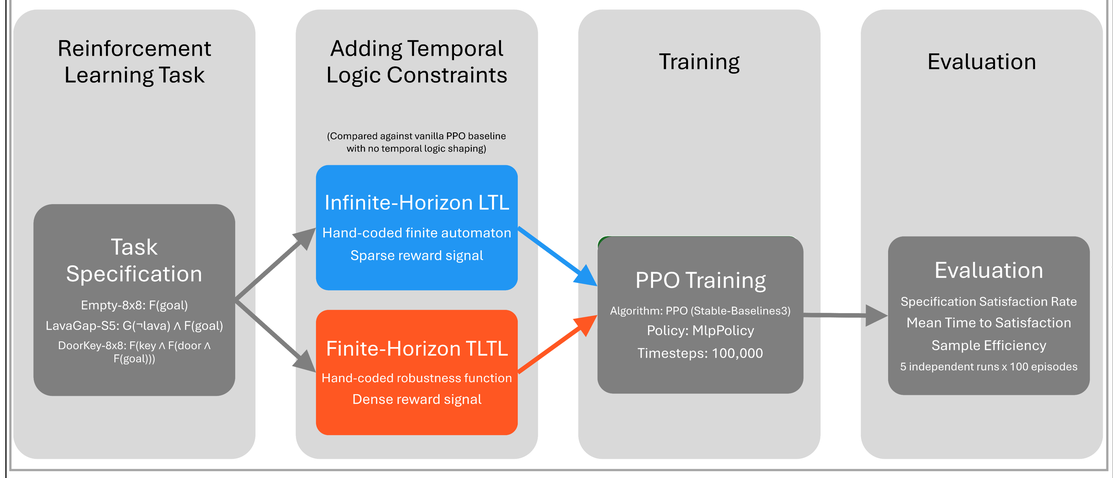
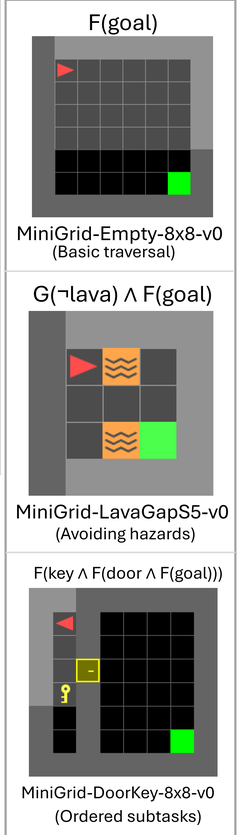


Figure 2



4. Conclusions

- **TLTL** robustness-based reward shaping **significantly outperformed** both baseline and infinite-horizon LTL on the sequential *DoorKey-8x8* task ($p<0.001$), achieving **94.8% satisfaction** vs **7.2%** and **0.6%** respectively.
- On simpler tasks (*Empty-8x8, LavaGap-S5*), all three methods performed comparably, suggesting temporal logic shaping provides the **greatest benefit as task complexity increases**.
- **Dense continuous reward signals (TLTL)** enable more *sample-efficient learning* for ordered subtask problems than **sparse automaton-based rewards (LTL).**
- LTL training showed **higher variance** across runs, suggesting *instability* even on simpler tasks.
- Results suggest **finite-horizon robustness-based specifications** are more practical for complex RL tasks, while *infinite-horizon methods* may offer theoretical guarantees at the cost of learning efficiency.

Figure 3

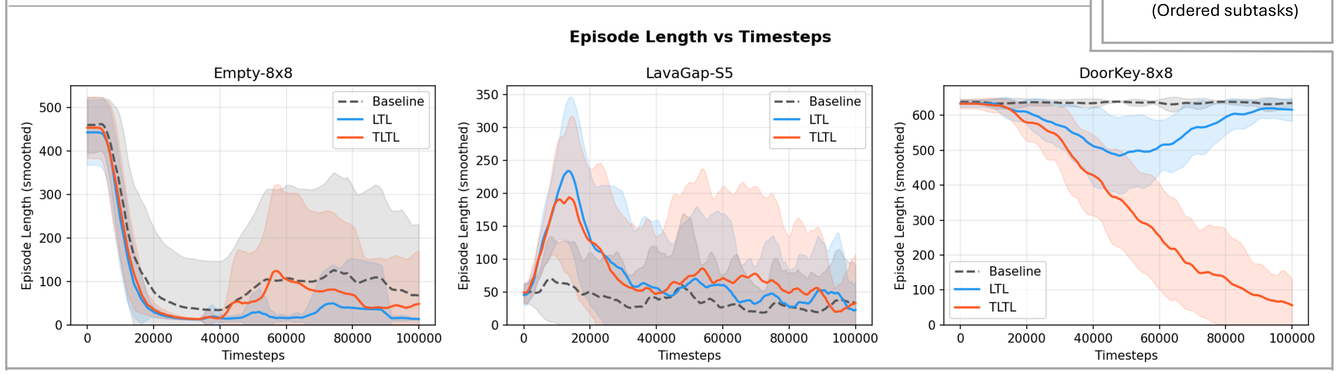


Figure 4

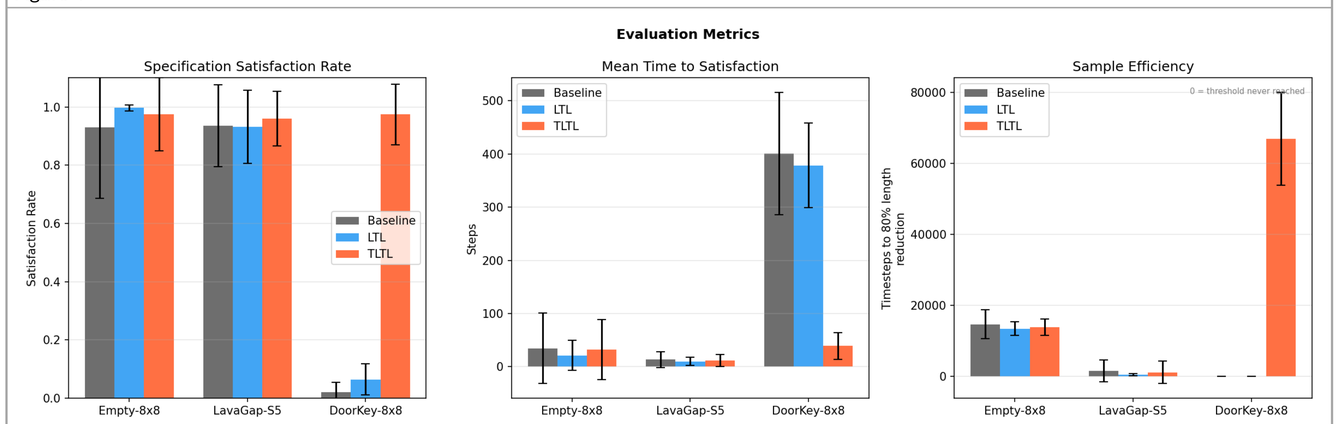
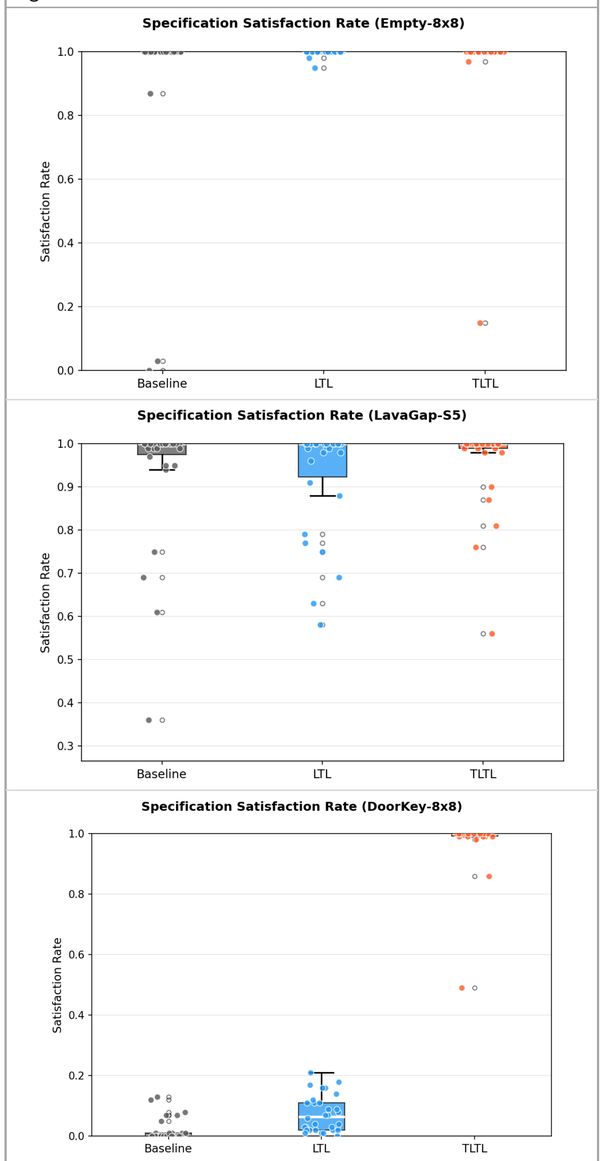


Figure 5



No significant differences between methods on Empty-8x8 and LavaGap-S5 ($p>0.05$). On **DoorKey-8x8**, all pairwise comparisons were **significant ($p<0.001$),** with **TLTL achieving nearly perfect satisfaction** while Baseline and LTL *largely failed*.

References

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